

HANFORD, CA, USA

WMO: 723898

Lat: 36.319N

Lon: 119.629W

Elev: 249

StdP: 14.56

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 53119

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.5	30.9	22.4	17.0	47.3	25.7	19.9	46.3	17.5	51.8	15.5	51.5	1.3	60	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.4	102.3	69.8	99.9	69.4	97.8	68.4	73.6	95.4	72.0	93.8	70.4	92.5	6.8	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	97.3	83.4	63.8	89.4	83.4	62.1	84.1	81.6	37.3	95.8	35.7	93.7	34.3	92.7	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.9	12.6	DB	25.0	106.7	2.9	2.3	22.9	108.4	21.2	109.7	19.6	111.0	17.5	112.6	(4)
(5)			WB	24.1	76.8	2.9	3.0	22.0	79.0	20.3	80.7	18.6	82.4	16.5	84.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	46.9	50.8	56.6	61.6	69.2	76.5	81.6	80.0	74.6	64.5	53.5	46.4		
	DBStd	13.45	5.08	4.86	5.89	6.64	6.16	5.87	4.37	3.99	5.52	5.65	5.58	5.19		
	HDD50	349	125	45	10	2	0	0	0	0	0	0	28	139		
	HDD65	2405	563	397	268	141	28	3	0	0	4	76	347	578		
	CDD50	5302	27	68	214	350	594	796	981	929	737	448	132	26		
	CDD65	1881	0	0	6	39	157	348	516	464	291	59	1	0		
	CDH74	27333	0	4	183	738	2300	5047	7495	6489	4004	1022	50	1		
	CDH80	15150	0	0	36	265	1039	2846	4581	3870	2170	340	3	0		
(14) Wind	WSAvg	5.2	3.2	4.0	5.1	6.5	7.4	7.6	6.6	6.2	5.2	4.0	3.1	3.1		
(15) Precipitation	PrecAvg	8.6	1.7	1.5	1.5	0.7	0.2	0.1	0.0	0.0	0.2	0.4	0.9	1.1		
	PrecMax	15.6	6.7	5.0	6.6	3.0	1.5	0.8	0.2	0.4	1.5	2.4	2.9	3.3		
	PrecMin	3.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	3.0	1.4	1.4	1.3	0.7	0.4	0.2	0.0	0.1	0.4	0.6	0.8	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.4	73.9	84.3	92.6	98.9	104.1	106.0	104.3	101.4	93.2	79.3	66.3		
		MCWB	53.2	58.2	61.5	64.0	69.3	70.8	70.8	69.8	68.2	66.2	60.3	55.4		
	2%	DB	63.3	69.4	79.0	87.3	93.9	100.3	102.6	101.0	98.1	87.9	74.4	63.0		
		MCWB	53.7	56.8	60.0	62.0	65.4	69.2	70.4	68.9	67.6	64.2	59.1	53.7		
	5%	DB	60.1	65.4	74.7	82.2	90.0	97.3	100.1	98.9	95.3	83.6	70.1	59.5		
		MCWB	52.7	55.1	58.6	60.3	63.9	67.5	70.3	68.8	66.5	61.9	57.2	52.0		
	10%	DB	56.7	62.2	70.7	77.5	86.0	93.7	97.7	96.4	92.0	80.2	65.8	56.5		
		MCWB	51.5	53.8	57.0	58.7	62.0	66.5	69.0	67.7	65.9	60.9	55.4	50.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.6	61.0	64.2	67.3	72.5	74.9	76.7	74.5	72.8	68.7	63.4	59.3		
		MCDB	62.2	68.0	77.6	86.5	91.8	97.7	98.3	95.0	92.0	85.9	73.5	63.2		
	2%	WB	55.8	58.5	62.1	63.9	68.1	72.3	74.1	72.3	70.0	66.3	60.7	56.1		
		MCDB	60.0	66.2	75.5	82.5	89.7	94.8	96.2	94.8	92.0	82.3	71.0	59.7		
	5%	WB	54.0	56.6	60.0	61.6	65.6	70.3	72.3	70.6	68.3	64.1	58.5	54.1		
		MCDB	57.7	63.8	72.2	78.9	85.7	92.0	94.1	93.3	90.9	79.8	67.8	57.8		
	10%	WB	51.9	54.6	57.9	59.6	63.4	68.3	70.7	69.2	66.8	62.2	56.5	51.6		
		MCDB	56.1	60.9	68.6	75.4	83.0	89.5	92.6	91.9	89.2	77.0	64.3	55.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.2	21.7	24.3	26.3	28.1	30.6	31.4	32.4	32.6	29.4	24.6	19.8		
		MCDBR	24.6	28.0	31.3	34.4	34.4	34.9	33.7	34.9	36.6	35.5	31.2	24.4		
	5% WB	MCWBR	15.7	16.3	14.5	13.2	12.0	10.9	10.6	10.7	12.0	14.3	16.5	15.4		
		MCWBR	18.5	23.8	28.1	31.8	31.5	31.4	30.5	32.3	33.5	30.5	27.2	16.9		
(40) Clear-Sky Solar Irradiance	taub		0.361	0.371	0.371	0.371	0.371	0.364	0.384	0.363	0.367	0.391	0.380	0.357		
	taud		2.313	2.304	2.320	2.346	2.367	2.386	2.358	2.393	2.392	2.272	2.252	2.329		
	Ebn at Noon		259	272	284	289	289	290	283	288	281	260	248	251		
	Edh at Noon		31	36	38	39	39	38	39	37	35	36	33	29		
(44) All-Sky Solar Radiation	RadAvg		774	1127	1609	2043	2417	2679	2562	2354	1967	1433	976	715		
	RadStd		111	124	133	144	124	81	88	63	54	73	54	96		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	DBAvg	+1.44	N/A	N/A	+1.60	N/A	N/A	-166	-506	+572
(47) Regional (27 neighbors)	DBAvg	+1.03	N/A	-1.98	+0.99	N/A	N/A	-49	-266	+340

Nomenclature: See separate page